

Rubber-Band Dynamics Comparison: Human-AI vs Human-Human vs AI-AI

Cross-Condition Valence Gap and Re-Alignment Analysis

PENPAL Project

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1 Overview

This notebook compares **rubber-band dynamics** across conditions using the exchange-aligned data structure.

The main quantity is the **valence gap**:

- `valence_gap = slot_1_valence - slot_2_valence`
- `gap_change = current_gap - previous_gap`

A negative slope in `gap_change ~ previous_gap` indicates a rubber-band effect: when the slots diverge, the next aligned exchange tends to pull them back together.

As in the updated valence notebook:

- `author_1 / author_2` are treated as **slot 1 / slot 2**
- the analysis is restricted to **complete exchanges**
- the temporal window uses `analysis_turn / turn_index`, not the old raw-turn logic

Primary cross-condition object. For each story i , fit the OLS

$$\text{gap_change}_{i,t} = \alpha_i + \beta_i \cdot \text{valence_gap_lag1}_{i,t} + \varepsilon_{i,t}$$

and define the per-story reversion strength as $R_i = -\beta_i$, so that larger positive R_i means a stronger pull back together. R_i is already label-invariant (the gap enters squared/reversal-style in the relationship). We

test $R_i \sim \text{condition}$ with ANOVA + Kruskal and pairwise BH-adjusted Wilcoxon. The pooled LMM slopes are kept as supplemental.

2 Load Data

Table 1: Loaded valence data for rubber-band analysis

condition	n_stories	n_turns
Human-AI	48	431
Human-Human	31	279
AI-AI	39	351

3 Preprocessing

Analysis rows: 943

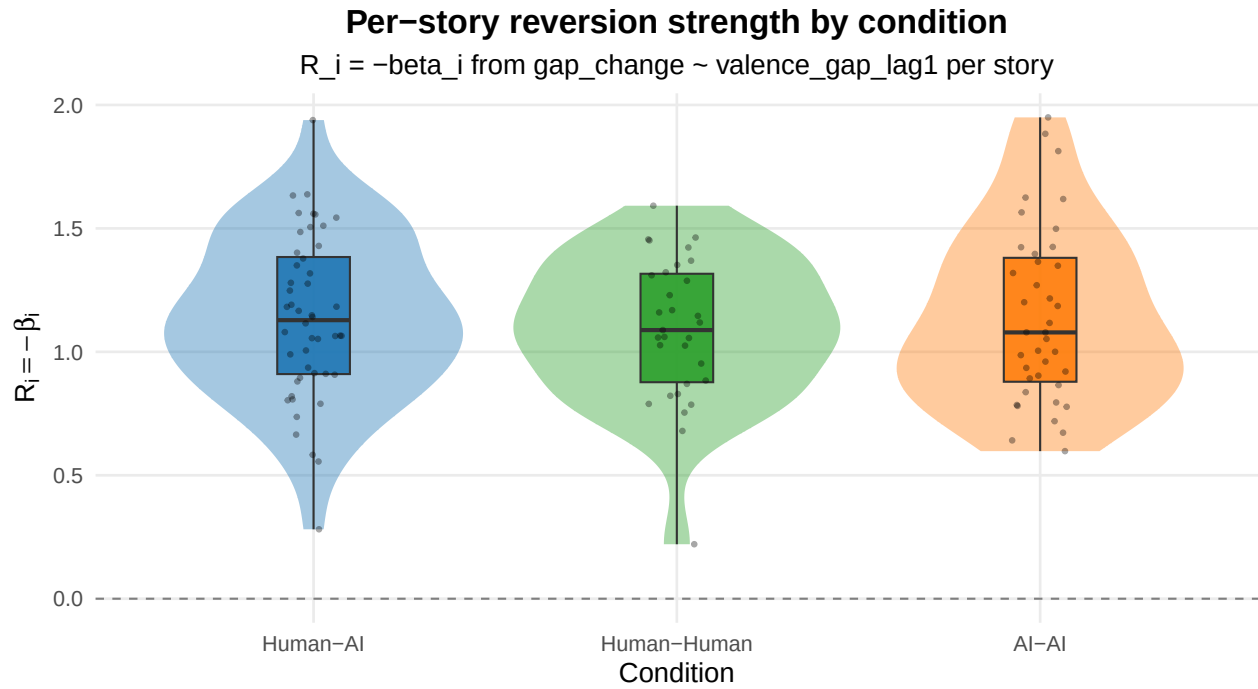
Stories: 118

4 Primary: Per-Story Reversion Strength

Table 2: Per-story reversion strength $R_i = -\beta$ by condition

condition	n	mean_R	median_R	sd_R
Human-AI	48	1.138	1.128	0.332
Human-Human	31	1.089	1.088	0.293
AI-AI	39	1.141	1.079	0.353

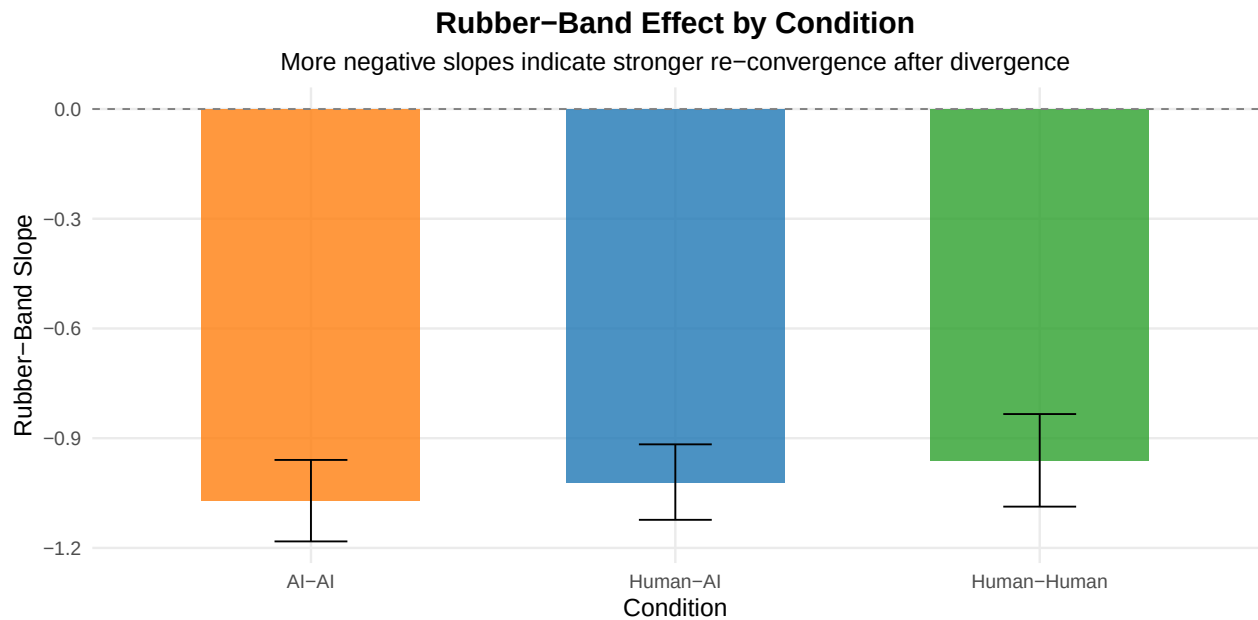
```
## One-way ANOVA: R ~ condition
##               Df Sum Sq Mean Sq F value Pr(>F)
## condition      2  0.058  0.02923   0.269   0.765
## Residuals    115 12.495  0.10866
##
## Kruskal-Wallis (robust): R ~ condition
##
## Kruskal-Wallis rank sum test
##
## data: R by condition
## Kruskal-Wallis chi-squared = 0.3273, df = 2, p-value = 0.849
##
## Pairwise Wilcoxon (BH-adjusted):
##
## Pairwise comparisons using Wilcoxon rank sum exact test
##
## data: per_story_rb$R and per_story_rb$condition
##
##           Human-AI Human-Human
## Human-Human 0.91      -
## AI-AI       0.91      0.91
##
## P value adjustment method: BH
```



5 Supplemental: Pooled Condition-Specific Slopes

Table 3: Rubber-band slopes by condition

condition	estimate	se	t	p	sig
Human-Human	-0.961	0.065	-14.874	0	***
Human-AI	-1.020	0.053	-19.375	0	***
AI-AI	-1.071	0.057	-18.844	0	***



5.1 Supplemental: Cross-Condition Interaction Model

```
## Cross-condition model summary:

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: gap_change ~ valence_gap_lag1 * condition + (1 | conversation_id)
## Data: df_analysis
##
## REML criterion at convergence: 999.1
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.07055 -0.64739 -0.00802  0.66504  3.02520
##
## Random effects:
## Groups           Name          Variance Std.Dev.
## conversation_id (Intercept) 0.006988 0.08359
## Residual                0.158941 0.39867
## Number of obs: 943, groups: conversation_id, 118
##
## Fixed effects:
##
##              Estimate Std. Error      df t value
## (Intercept)    -0.09761    0.02412  91.80317   -4.047
## valence_gap_lag1 -0.95972    0.04965 919.56993  -19.331
## conditionHuman-Human    0.10678    0.03805  88.10540    2.806
## conditionAI-AI         0.09020    0.03565  88.49955    2.530
## valence_gap_lag1:conditionHuman-Human -0.02791    0.07694 934.58229   -0.363
## valence_gap_lag1:conditionAI-AI    -0.12876    0.08515 928.16834   -1.512
##
##              Pr(>|t|)
## (Intercept)    0.000108 ***
## valence_gap_lag1    < 2e-16 ***
## conditionHuman-Human    0.006170 **
## conditionAI-AI         0.013161 *
## valence_gap_lag1:conditionHuman-Human 0.716878
## valence_gap_lag1:conditionAI-AI    0.130829
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) vln__1 cndH-H cAI-AI v__1:H
## vlnc_gp_lg1    0.190
## cndtnHmn-Hm   -0.634 -0.120
## condtnAI-AI   -0.677 -0.128  0.429
## vlnc__1:H-H   -0.123 -0.645  0.074  0.083
## vl__1:AI-AI   -0.111 -0.583  0.070  0.090  0.376
##
##
## Condition-specific slopes:
## condition valence_gap_lag1.trend SE df lower.CL upper.CL
## Human-AI    -0.960 0.0498 924    -1.06   -0.862
## Human-Human  -0.988 0.0589 937    -1.10   -0.872
## AI-AI        -1.088 0.0693 906    -1.22   -0.953
##
## Degrees-of-freedom method: kenward-roger
```

```

## Confidence level used: 0.95
##
## Pairwise slope contrasts:
## contrast estimate SE df t.ratio p.value
## (Human-AI) - (Human-Human) 0.0279 0.0771 935 0.362 1.0000
## (Human-AI) - (AI-AI) 0.1288 0.0853 931 1.510 0.3944
## (Human-Human) - (AI-AI) 0.1009 0.0909 925 1.109 0.8028
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: bonferroni method for 3 tests

```

6 Supplemental: General Gap Structure

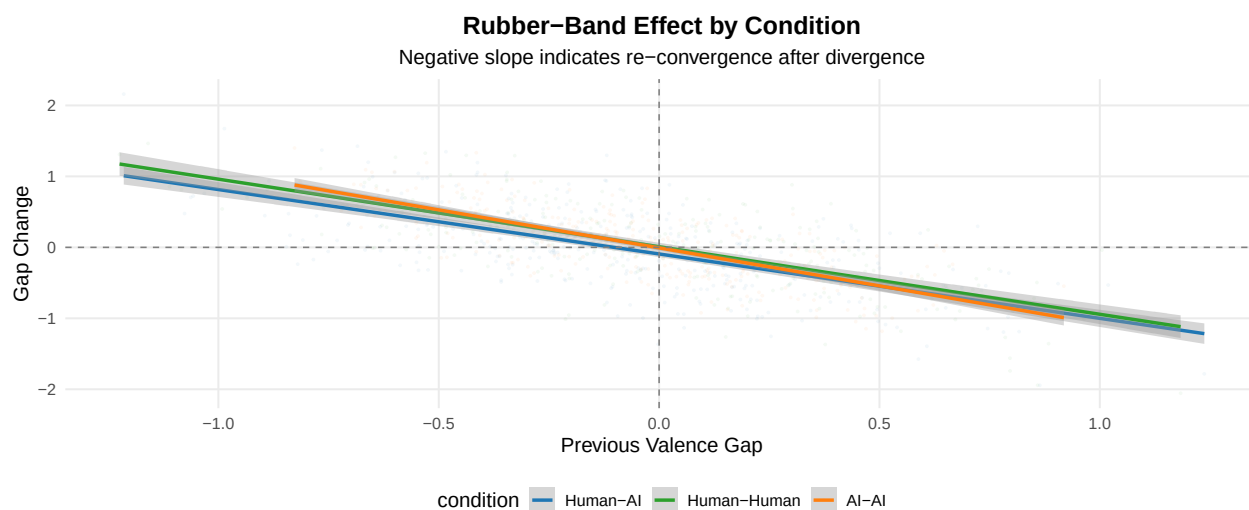
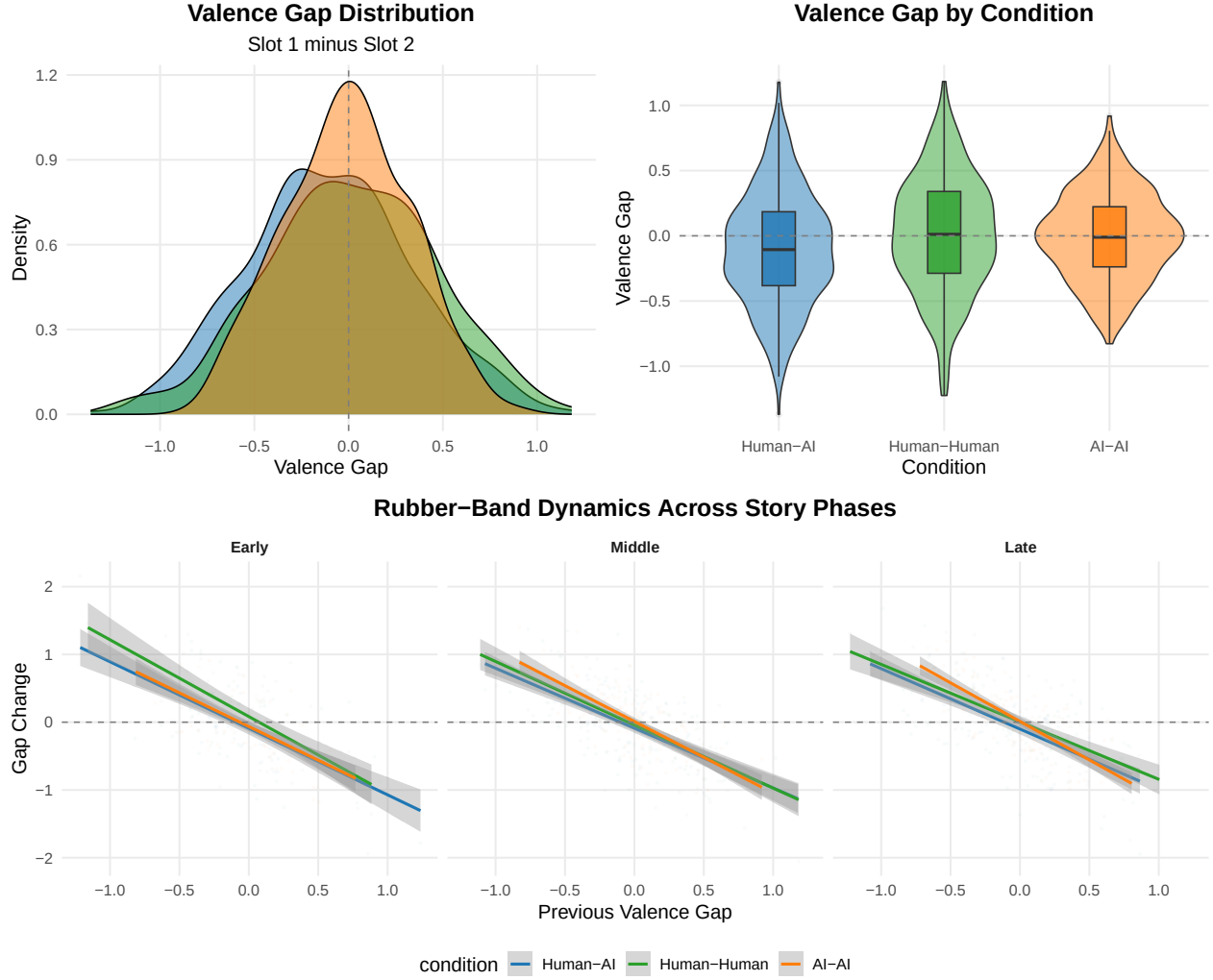


Table 4: Gap and gap-change descriptives by condition

condition	n_turns	mean_gap	sd_gap	mean_gap_change	sd_gap_change	rubber_band_r
Human-AI	383	-0.102	0.436	-0.009	0.579	-0.662
Human-Human	248	0.009	0.446	0.006	0.612	-0.685
AI-AI	312	-0.007	0.331	0.003	0.484	-0.731



7 Summary

Primary cross-condition result: per-story reversion strength $R_i = -\beta_i$ from `gap_change ~ valence_gap_lag1`, tested with ANOVA + Kruskal and pairwise BH-adjusted Wilcoxon.

Supplemental views retained:

- pooled within-condition LMM slopes (population-level rubber-band estimate per condition)
- cross-condition interaction LMM (`gap_change ~ valence_gap_lag1 * condition + (1|story)`)
- gap distributions and story-phase splits

Deferred to a follow-up pass:

- starter-mechanism analysis of the reversion object
- Human-AI decomposition of who pulls the gap back (requires per-turn speaker identity)

Analysis completed: 2026-04-23 12:22:11.876419